



December 18, 2018

Micro Therapeutics, Inc. d/b/a ev3 Neurovascular
Nancy Drew, PhD
Senior Regulatory Affairs Specialist
9775 Toledo Way
Irvine, California 92618

Re: K183185

Trade/Device Name: Riptide™ Aspiration System (Riptide™ Large Bore Aspiration Tubing)
Regulation Number: 21 CFR 870.1250
Regulation Name: Percutaneous Catheter
Regulatory Class: Class II
Product Code: NRY
Dated: November 16, 2018
Received: November 19, 2018

Dear Nancy Drew, PhD:

We have reviewed your Section 510(k) premarket notification of intent to market the device referenced above and have determined the device is substantially equivalent (for the indications for use stated in the enclosure) to legally marketed predicate devices marketed in interstate commerce prior to May 28, 1976, the enactment date of the Medical Device Amendments, or to devices that have been reclassified in accordance with the provisions of the Federal Food, Drug, and Cosmetic Act (Act) that do not require approval of a premarket approval application (PMA). You may, therefore, market the device, subject to the general controls provisions of the Act. Although this letter refers to your product as a device, please be aware that some cleared products may instead be combination products. The 510(k) Premarket Notification Database located at <https://www.accessdata.fda.gov/scripts/cdrh/cfdocs/cfpmn/pmn.cfm> identifies combination product submissions. The general controls provisions of the Act include requirements for annual registration, listing of devices, good manufacturing practice, labeling, and prohibitions against misbranding and adulteration. Please note: CDRH does not evaluate information related to contract liability warranties. We remind you, however, that device labeling must be truthful and not misleading.

If your device is classified (see above) into either class II (Special Controls) or class III (PMA), it may be subject to additional controls. Existing major regulations affecting your device can be found in the Code of Federal Regulations, Title 21, Parts 800 to 898. In addition, FDA may publish further announcements concerning your device in the Federal Register.

Please be advised that FDA's issuance of a substantial equivalence determination does not mean that FDA has made a determination that your device complies with other requirements of the Act or any Federal statutes and regulations administered by other Federal agencies. You must comply with all the Act's

requirements, including, but not limited to: registration and listing (21 CFR Part 807); labeling (21 CFR Part 801); medical device reporting (reporting of medical device-related adverse events) (21 CFR 803) for devices or postmarketing safety reporting (21 CFR 4, Subpart B) for combination products (see <https://www.fda.gov/CombinationProducts/GuidanceRegulatoryInformation/ucm597488.htm>); good manufacturing practice requirements as set forth in the quality systems (QS) regulation (21 CFR Part 820) for devices or current good manufacturing practices (21 CFR 4, Subpart A) for combination products; and, if applicable, the electronic product radiation control provisions (Sections 531-542 of the Act); 21 CFR 1000-1050.

Also, please note the regulation entitled, "Misbranding by reference to premarket notification" (21 CFR Part 807.97). For questions regarding the reporting of adverse events under the MDR regulation (21 CFR Part 803), please go to <http://www.fda.gov/MedicalDevices/Safety/ReportaProblem/default.htm>.

For comprehensive regulatory information about medical devices and radiation-emitting products, including information about labeling regulations, please see Device Advice (<https://www.fda.gov/MedicalDevices/DeviceRegulationandGuidance/>) and CDRH Learn (<http://www.fda.gov/Training/CDRHLearn>). Additionally, you may contact the Division of Industry and Consumer Education (DICE) to ask a question about a specific regulatory topic. See the DICE website (<http://www.fda.gov/DICE>) for more information or contact DICE by email (DICE@fda.hhs.gov) or phone (1-800-638-2041 or 301-796-7100).

Sincerely,


Xiaolin Zheng -S

for Carlos L. Peña, PhD, MS
Director
Division of Neurological
and Physical Medicine Devices
Office of Device Evaluation
Center for Devices and Radiological Health

Enclosure

Indications for Use

510(k) Number (if known)

K183185

Device Name

Riptide™ Aspiration System (Riptide™ Large Bore Aspiration Tubing)

Indications for Use (Describe)

The Riptide™ Aspiration System is intended for use in the revascularization of patients with acute ischemic stroke secondary to intracranial large vessel occlusive disease (within the internal carotid, middle cerebral M1 and M2 segments, basilar, and vertebral arteries) within 8 hours of symptom onset. Patients who are ineligible for intravenous tissue plasminogen activator (IV t-PA) or who fail IV t-PA therapy are candidates for treatment.

Type of Use (Select one or both, as applicable)

☒ Prescription Use (Part 21 CFR 801 Subpart D)

☐ Over-The-Counter Use (21 CFR 801 Subpart C)

CONTINUE ON A SEPARATE PAGE IF NEEDED.

This section applies only to requirements of the Paperwork Reduction Act of 1995.

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510(k) Summary: K183185

510(k) Owner:	Micro Therapeutics, Inc. d/b/a ev3 Neurovascular 9775 Toledo Way Irvine, CA 92618 Establishment Registration: 2029214
Contact Person:	Nancy Drew Senior Regulatory Affairs Specialist Telephone: (949) 297-9563 Email: nancy.k.drew@medtronic.com

Date Summary Prepared:	December 13, 2018
Trade Name of Device:	Riptide™ Aspiration System (Riptide™ Large Bore Aspiration Tubing)
Common Name of Device:	Catheter, Thrombus Retriever
Review Panel:	Neurology
Product Code:	NRV
Regulation Number:	21 CFR 870.1250
Regulation Name:	Percutaneous Catheter
Device Classification	Class II
Predicate Device:	Riptide™ Aspiration System (Riptide™ Aspiration Tubing) 510(k): K172448

Device Description:

The Riptide™ Large Bore Aspiration Tubing is provided sterile and intended for single-use. The Riptide™ Large Bore Aspiration Tubing serves as a conduit to supply vacuum from the Riptide™ Aspiration Pump to the distal tip of the catheter. The Riptide™ Large Bore Aspiration Tubing provides a connection between the sterile and non-sterile environments. The proximal end of the Riptide™ Large Bore Aspiration Tubing is connected to the Riptide™ Collection Canister (outside of the sterile environment) while the distal end of the Riptide™ Large Bore Aspiration Tubing is connected to the catheter (inside the sterile environment). The Riptide™ Collection Canister is connected to the Riptide™ Aspiration Pump (also outside of the sterile environment) via the Intermediate Tubing. The Riptide™ Large Bore Aspiration Tubing is designed to allow the user to control the application of aspiration via an in-line flow switch.

Indication for Use Statement:

The Riptide™ Aspiration System is intended for use in the revascularization of patients with acute ischemic stroke secondary to intracranial large vessel occlusive disease (within the internal carotid, middle cerebral M1 and M2 segments, basilar, and vertebral arteries) within 8 hours of symptom onset. Patients who are ineligible for intravenous tissue plasminogen activator (IV t-PA) or who fail IV t-PA therapy are candidates for treatment.

Device Comparison:

	Predicate: Riptide™ Aspiration Tubing (K172448)	Subject: Riptide™ Large Bore Aspiration Tubing
Indication for Use (IFU) Statement	The Riptide™ Aspiration System is intended for use in the revascularization of patients with acute ischemic stroke secondary to intracranial large vessel occlusive disease (within the internal carotid, middle cerebral M1 and M2 segments, basilar, and vertebral arteries) within 8 hours of symptom onset. Patients who are ineligible for intravenous tissue plasminogen activator (IV t-PA) or who fail IV t-PA therapy are candidates for treatment	Same
Materials	Biocompatible, commonly utilized for interventional devices.	Same
<i>Dimensions</i>		
Usable Length	112.0"	Same
Distal Length	7.0"	Same
Inner Diameter	0.088"	0.110"
Outer Diameter	0.188"	Same
<i>Sterilization</i>		
Method	Ethylene Oxide (EO)	Same
<i>Packaging</i>		
Shelf Life	36 Months	Same

Biocompatibility:

Biocompatibility was leveraged for the Riptide™ Large Bore Aspiration Tubing from the predicate device.

Performance Data – Bench:

Non-clinical bench testing was leveraged and conducted to evaluate the performance of the Riptide™ Large Bore Aspiration Tubing.

The following non-clinical bench testing was leveraged for the Riptide™ Large Bore Aspiration Tubing:

Test	Test Method Summary	Results
<i>Performance</i>		
Compatibility	Leveraged from predicate device	The Riptide™ Large Bore Aspiration Tubing met the acceptance criteria for compatibility with the Arc™ Catheter, React™ 68 Catheter, React™ 71 Catheter, and Riptide™ Collection Canister
<i>Microbial</i>		

Test	Test Method Summary	Results
Ethylene Oxide (EO) Residual	Leveraged from predicate device	The Riptide™ Large Bore Aspiration Tubing met the acceptance criteria for ethylene oxide residual.
Ethylene Chlorohydrin (ECH) Residual	Leveraged from predicate device	The Riptide™ Large Bore Aspiration Tubing met the acceptance criteria for ethylene chlorohydrin residual.
Bioburden Recovery	Leveraged from predicate device	The Riptide™ Large Bore Aspiration Tubing met the acceptance criteria for bioburden recovery.
Bioburden	Leveraged from predicate device	The Riptide™ Large Bore Aspiration Tubing met the acceptance criteria for bioburden.
Bacterial Endotoxin	Leveraged from predicate device	The Riptide™ Large Bore Aspiration Tubing met the acceptance criteria for bacterial endotoxin.
<i>Packaging</i>		
Aseptic Presentation	Leveraged from predicate device	The Riptide™ Large Bore Aspiration Tubing met the acceptance criteria for aseptic presentation.
Seal Width	Leveraged from predicate device	The Riptide™ Large Bore Aspiration Tubing met the acceptance criteria for seal width.
Dye Leak	Leveraged from predicate device	The Riptide™ Large Bore Aspiration Tubing met the acceptance criteria for dye leak.
Visual Inspection	Leveraged from predicate device	The Riptide™ Large Bore Aspiration Tubing met the acceptance criteria for visual inspection.
Bubble Leak	Leveraged from predicate device	The Riptide™ Large Bore Aspiration Tubing met the acceptance criteria for bubble leak.
Legibility	Leveraged from predicate device	The Riptide™ Large Bore Aspiration Tubing met the acceptance criteria for labeling legibility.
Foreign Material	Leveraged from predicate device	The Riptide™ Large Bore Aspiration Tubing met the acceptance criteria for foreign material.

Test	Test Method Summary	Results
Seal Strength	Leveraged from predicate device	The Riptide™ Large Bore Aspiration Tubing met the acceptance criteria for seal strength.

The following non-clinical bench testing was conducted:

Test	Test Method Summary	Results
Dimensional Inspection	The overall and distal length, proximal and distal inner and outer diameter of the Riptide™ Large Bore Aspiration Tubing is measured.	The Riptide™ Large Bore Aspiration Tubing met the acceptance criteria for dimensional inspection.
Identification (ON/OFF Heat Shrink Bands)	The heat shrink bands of the Riptide™ Large Bore Aspiration Tubing is visually inspected.	The Riptide™ Large Bore Aspiration Tubing met the acceptance criteria for identification of the heat shrink bands.
Identification (Flow Switch)	The flow switch of the Riptide™ Large Bore Aspiration Tubing is visually inspected.	The Riptide™ Large Bore Aspiration Tubing met the acceptance criteria for identification of the flow switch.
Joint Tensile Strength	The Riptide™ Large Bore Aspiration Tubing is affixed to upper and lower grips of the tensile test fixture and stretched. The maximum value before failure is recorded.	The Riptide™ Large Bore Aspiration Tubing met the acceptance criteria for joint tensile strength.
Leak	The Riptide™ Large Bore Aspiration Tubing is pressurized to specified vacuum pressure and any decay in pressure is recorded.	The Riptide™ Large Bore Aspiration Tubing met the acceptance criteria for leak (vacuum decay).
Degree of Collapse	Degree of Collapse for the Riptide™ Large Bore Aspiration Tubing is performed per ISO 10079-3.	The Riptide™ Large Bore Aspiration Tubing met the acceptance criteria for degree of collapse.

In addition, the following non-clinical bench testing was leveraged to evaluate the performance of Riptide™ Aspiration System (Riptide™ Large Bore Aspiration Tubing):

Test	Test Method Summary	Results
Recanalization	Leveraged from predicate device	The Riptide™ Aspiration System (Riptide™ Large Bore Aspiration Tubing) met the acceptance criteria for recanalization.

Test	Test Method Summary	Results
Vacuum Pressure	Leveraged from predicate device	The Riptide™ Aspiration System (Riptide™ Large Bore Aspiration Tubing) met the acceptance criteria for vacuum pressure.
Volumetric Flow Rate	Leveraged from predicate device	The Riptide™ Aspiration System (Riptide™ Large Bore Aspiration Tubing) met the acceptance criteria for volumetric flow rate.
Usability	Leveraged from predicate device	The Riptide™ Aspiration System (Riptide™ Large Bore Aspiration Tubing) met the acceptance criteria for usability.

Performance Data – Animal:

A determination of substantial equivalence is based upon non-clinical bench testing as there is no change to the indication for use statement, fundamental scientific technology, or materials of construction.

Performance Data – Clinical:

A determination of substantial equivalence is based upon non-clinical bench testing as there is no change to the indication for use statement, fundamental scientific technology, or materials of construction.

Conclusion:

The design modification incorporated into the Riptide™ Large Bore Aspiration Tubing do not alter the indication for use statement, fundamental scientific technology, or materials of construction. In addition, the design modification does not raise new questions on the safety and effectiveness of the device. Therefore, the Riptide™ Large Bore Aspiration Tubing is substantially equivalent to the predicate device.